

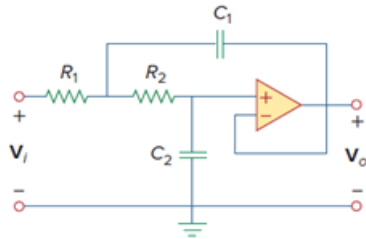
Problem

A second-order active filter known as a Butterworth filter is shown in Fig. 14.95.

(a) Find the transfer function V_o/V_i .

(b) Show that it is a low-pass filter.

Figure 14.95:



Step-by-step solution

Step 1 of 2

(a)

Refer to the Figure 14.95 in the text book.

Assume the node voltage V_x between resistors R_1 and R_2 .

Apply Kirchhoff's current law at noninverting terminal.

$$\frac{V_0 - V_x}{R_2} + \frac{V_0}{\left(\frac{1}{sC_2}\right)} = 0$$

$$V_0 \left[\frac{1}{R_2} + sC_2 \right] = \frac{V_x}{R_2}$$

$$V_x = V_0 [1 + sC_2 R_2]$$

$$V_x = V_0 \left[1 + \frac{sC_2}{G_2} \right]$$

Apply Kirchhoff's current law at V_x .

$$\frac{V_x - V_0}{\left(\frac{1}{sC_1}\right)} + \frac{V_x - V_0}{R_2} + \frac{V_x - V_{in}}{R_1} = 0$$

$$V_x \left[sC_1 + \frac{1}{R_1} + \frac{1}{R_2} \right] - V_0 \left[sC_1 + \frac{1}{R_2} \right] = \frac{V_{in}}{R_1}$$

$$V_x [sC_1 + G_1 + G_2] - V_0 [sC_1 + G_2] = G_1 V_{in}$$

Substitute $V_0 \left[1 + \frac{sC_2}{G_2} \right]$ for V_x .

$$V_0 \left[1 + \frac{sC_2}{G_2} \right] [sC_1 + G_1 + G_2] - V_0 [sC_1 + G_2] = G_1 V_{in}$$

$$V_0 \left[\left[1 + \frac{sC_2}{G_2} \right] [sC_1 + G_1 + G_2] - [sC_1 + G_2] \right] = G_1 V_{in}$$

$$V_0 \left[\frac{G_1 G_2 + sC_2 (G_1 + G_2 + sC_1)}{G_2} \right] = G_1 V_{in}$$

$$V_0 [G_1 G_2 + sC_2 (G_1 + G_2 + sC_1)] = G_1 G_2 V_{in}$$

$$\frac{V_0}{V_{in}} = \frac{G_1 G_2}{[G_1 G_2 + sC_2 (G_1 + G_2 + sC_1)]}$$

Therefore, the expression of transfer function is

$$\boxed{\frac{G_1 G_2}{G_1 G_2 + sC_2 (G_1 + G_2 + sC_1)}}$$

Step 2 of 2

(b)

Determine the system response at low frequency.

Substitute $\omega = 0$ in transfer function.

$$\begin{aligned}\mathbf{H}(0) &= \frac{G_1 G_2}{G_1 G_2 + (0) C_2 (G_1 + G_2 + (0) C_1)} \\ &= \frac{G_1 G_2}{G_1 G_2 + 0} \\ &= \frac{G_1 G_2}{G_1 G_2} \\ &= 1\end{aligned}$$

Determine the system response at high frequency.

Substitute $\omega = \infty$ in transfer function.

$$\begin{aligned}\mathbf{H}(\infty) &= \frac{G_1 G_2}{G_1 G_2 + (\infty) C_2 (G_1 + G_2 + (\infty) C_1)} \\ &= \frac{G_1 G_2}{\infty} \\ &= 0\end{aligned}$$

The system allows at low frequency signals and attenuates high frequency signals.

Hence, the filter is **low pass filter**.